

# CSE4227 Digital Image Processing

## Chapter 03 – Sharpening Spatial Filter

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# Today's Contents

❑ Sharpening spatial filter

❑ Derivatives of Image

- 1<sup>st</sup> derivative
- 2<sup>nd</sup> derivative

❑ Laplacian Filter

❑ Laplacian Image Enhancement

❑ Gradient Operators

❑ Difference filters

❑ Combining filtering techniques

■ Chapter 3 from R.C. Gonzalez and R.E. Woods, Digital Image Processing (3rd Edition), Prentice Hall, 2008 [ **Section 3.6, 3.7** ]

# Sharpening Spatial Filters

Previously we have looked at smoothing filters which **remove fine detail**.

*Sharpening spatial filters* seek to **highlight fine detail**.

- Remove blurring from images
- Highlight edges
- Useful for emphasizing transitions in image intensity

# Sharpening Spatial Filters

## Some Applications

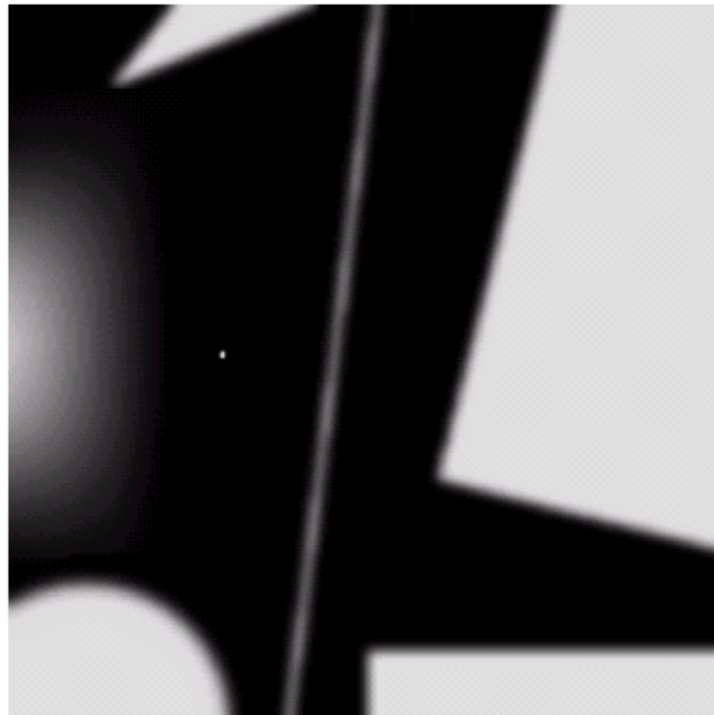
- ☐ Photo Enhancement
- ☐ Medical image visualization
- ☐ Industrial defect detection
- ☐ Electronic printing
- ☐ Autonomous guidance in military systems

# Spatial Differentiation

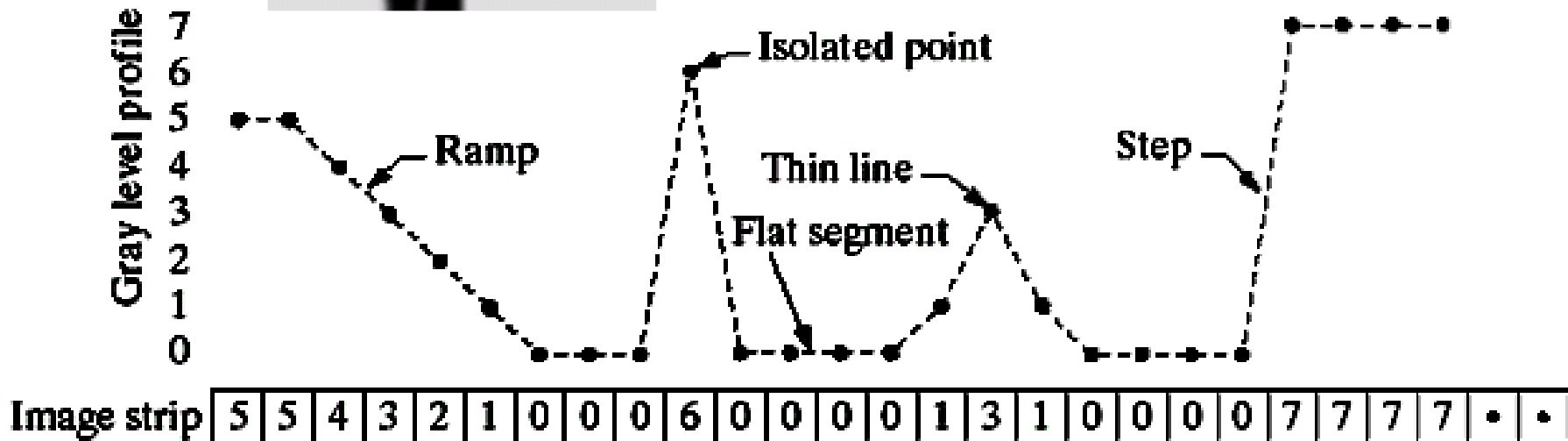
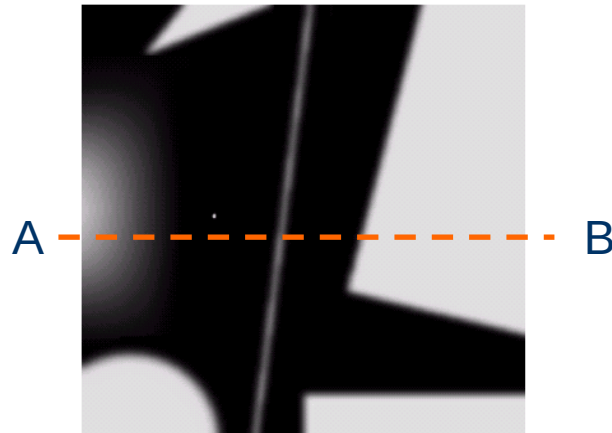
- Sharpening filters are based on **first- and second-order derivatives of image.**
- The derivatives of a digital function are defined **in terms of differences.**
- ***i.e.*** Sharpening filters are based on ***spatial differentiation***
- Differentiation measures **the rate of change of a function.**

# Spatial Differentiation

- ◆ Let's consider a simple 1 dimensional example



# Spatial Differentiation



# 1<sup>st</sup> Derivative in Digital Form

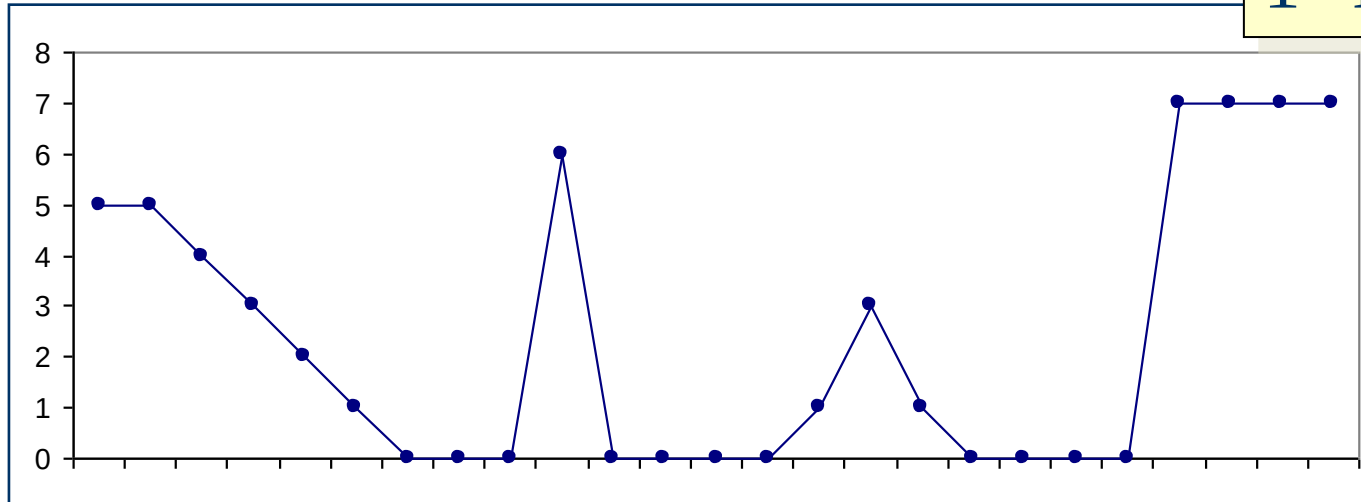
- The 1<sup>st</sup> derivative of a function is given by:

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

- Its just the difference between subsequent values and measures the rate of change of the function

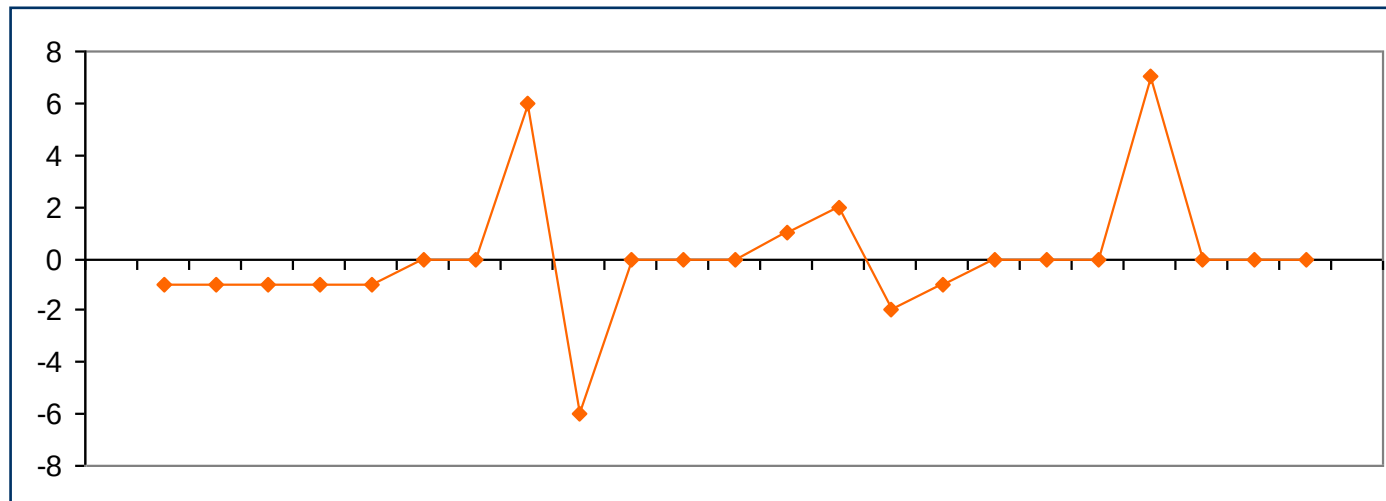


# 1<sup>st</sup> Derivative



|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5 | 5 | 4 | 3 | 2 | 1 | 0 | 0 | 0 | 6 | 0 | 0 | 0 | 0 | 1 | 3 | 1 | 0 | 0 | 0 | 0 | 7 | 7 | 7 | 7 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

|  |    |    |    |    |    |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |   |   |  |
|--|----|----|----|----|----|---|---|---|----|---|---|---|---|---|----|----|---|---|---|---|---|---|---|--|
|  | -1 | -1 | -1 | -1 | -1 | 0 | 0 | 6 | -6 | 0 | 0 | 0 | 1 | 2 | -2 | -1 | 0 | 0 | 0 | 7 | 0 | 0 | 0 |  |
|--|----|----|----|----|----|---|---|---|----|---|---|---|---|---|----|----|---|---|---|---|---|---|---|--|



Derivative is nonzero along the entire ramp, zero in flat area,.

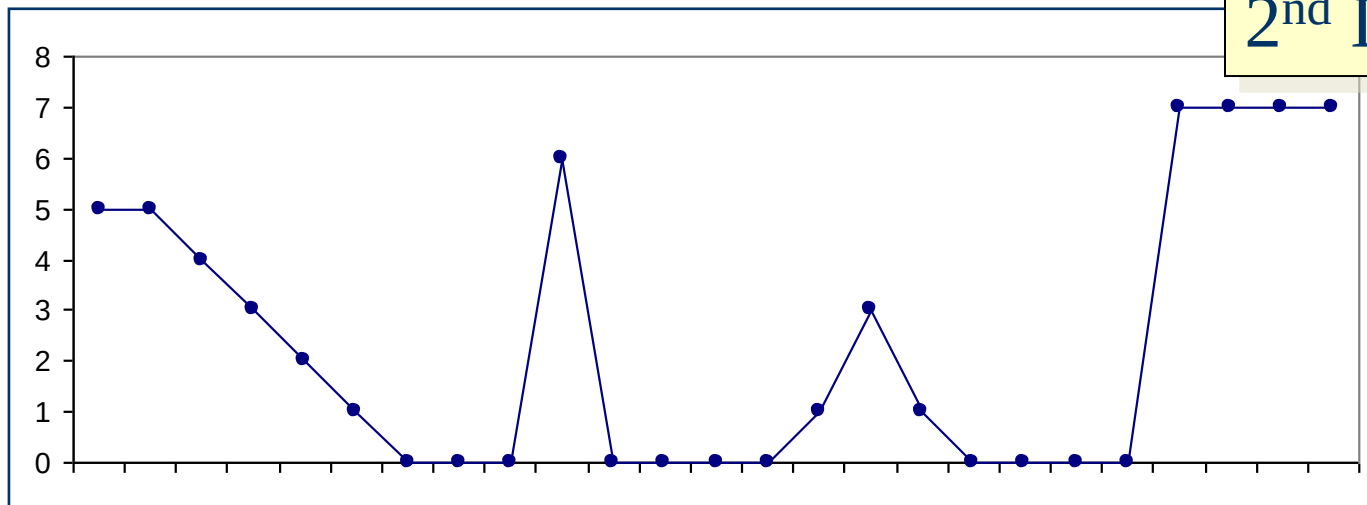
# 2<sup>nd</sup> Derivative in Digital Form

The 2nd derivative of a function is given by:

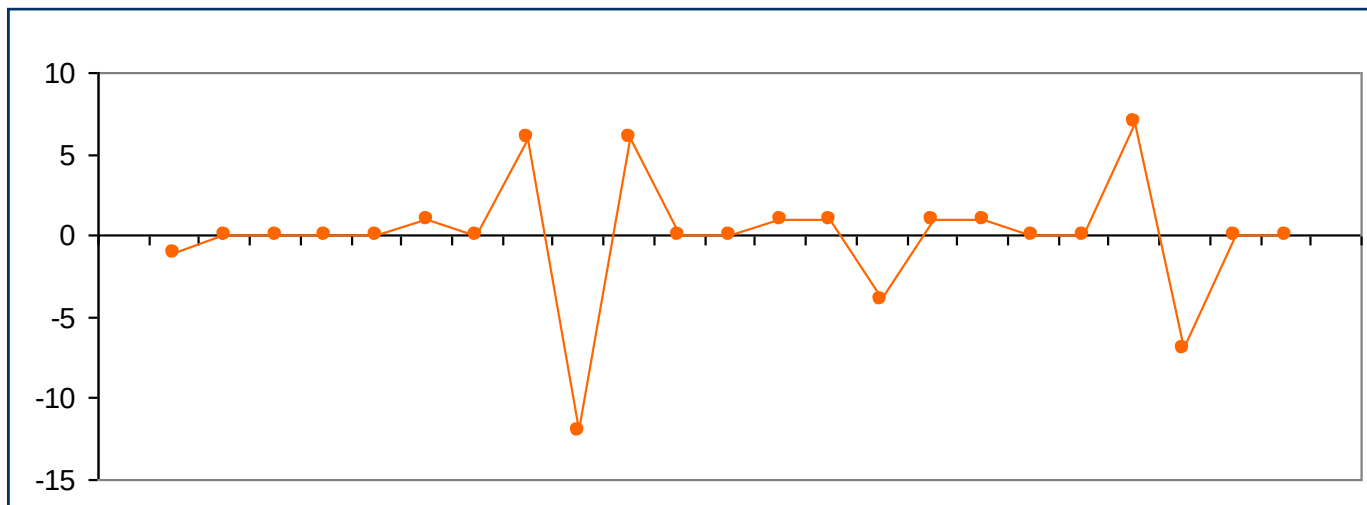
$$\frac{\partial^2 f}{\partial^2 x} = f(x+1) + f(x-1) - 2f(x)$$

Simply takes into account the values both before (backward) and after (forward) the current value

## 2<sup>nd</sup> Derivative



|    |   |   |   |   |   |   |   |     |   |   |   |   |   |    |   |   |   |   |   |    |   |   |   |   |
|----|---|---|---|---|---|---|---|-----|---|---|---|---|---|----|---|---|---|---|---|----|---|---|---|---|
| 5  | 5 | 4 | 3 | 2 | 1 | 0 | 0 | 0   | 6 | 0 | 0 | 0 | 0 | 1  | 3 | 1 | 0 | 0 | 0 | 0  | 7 | 7 | 7 | 7 |
| -1 | 0 | 0 | 0 | 0 | 1 | 0 | 6 | -12 | 6 | 0 | 0 | 1 | 1 | -4 | 1 | 1 | 0 | 0 | 7 | -7 | 0 | 0 |   |   |



Derivative is nonzero at the onset and end of ramp, stronger response at and around the point.

# Sharpening Spatial Filters

## 1. LAPLACIAN

- Use of 2<sup>nd</sup> Derivative for Image Enhancement

## 2. SOBEL (Gradient Operators)

- Use of 1st Derivative for Image Enhancement

# Use of 2<sup>nd</sup> Derivative for Image Enhancement

The 2nd derivative is more useful for image enhancement than the 1st derivative - *Stronger response to fine detail*

The first sharpening filter we will look at is the ***Laplacian***

# 2nd derivatives for image Sharpening - For Two Dimensional

□ 2-D 2<sup>nd</sup> derivatives => Laplacian

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

=>discrete formulation

$$\begin{aligned}\nabla^2 f &= [f(x+1, y) + f(x-1, y) - 2f(x, y)] \\ &\quad + [f(x, y+1) + f(x, y-1) - 2f(x, y)] \\ &= [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x, y)\end{aligned}$$

# 2<sup>nd</sup> Derivative in Two Dimension

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x) \quad \longrightarrow \quad \begin{bmatrix} 1 & -2 & 1 \end{bmatrix}$$

x kernel

$$\frac{\partial^2 f}{\partial y^2} = f(y+1) + f(y-1) - 2f(y) \quad \longrightarrow \quad \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

y kernel

|   |    |   |
|---|----|---|
| 0 | 0  | 0 |
| 1 | -2 | 1 |
| 0 | 0  | 0 |

 + 

|   |    |   |
|---|----|---|
| 0 | 1  | 0 |
| 0 | -2 | 0 |
| 0 | 1  | 0 |

 = 

|   |    |   |
|---|----|---|
| 0 | 1  | 0 |
| 1 | -4 | 1 |
| 0 | 1  | 0 |

# 1. Laplacian Filter

So, the Laplacian can be given as follows:

$$\nabla^2 f = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x, y)$$

We can implement it using this filter.

|   |    |   |
|---|----|---|
| 0 | 1  | 0 |
| 1 | -4 | 1 |
| 0 | 1  | 0 |



# Types of Laplacian Kernels

|   |    |   |   |    |   |
|---|----|---|---|----|---|
| 0 | 1  | 0 | 1 | 1  | 1 |
| 1 | -4 | 1 | 1 | -8 | 1 |
| 0 | 1  | 0 | 1 | 1  | 1 |

|    |    |    |    |    |    |
|----|----|----|----|----|----|
| 0  | -1 | 0  | -1 | -1 | -1 |
| -1 | 4  | -1 | -1 | 8  | -1 |
| 0  | -1 | 0  | -1 | -1 | -1 |

|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 3.39**

(a) Filter mask used to implement the digital Laplacian, as defined in Eq. (3.7-4).

(b) Mask used to implement an extension of this equation that includes the diagonal neighbors. (c) and (d) Two other implementations of the Laplacian.

# Laplacian Image Enhancement

Another Example:

|    |    |    |
|----|----|----|
| 0  | -1 | 0  |
| -1 | 4  | -1 |
| 0  | -1 | 0  |



**Original Image**

+



**Laplace Sharpened  
image**

=



**Laplace filtered  
image**

# Laplacian Filter

**Example:** apply the following Laplacian filter on the highlighted and underlined pixel

|    |          |    |
|----|----------|----|
| 0  | -1       | 0  |
| -1 | <u>4</u> | -1 |
| 0  | -1       | 0  |

|     |     |            |     |     |
|-----|-----|------------|-----|-----|
| 153 | 157 | 156        | 153 | 155 |
| 159 | 156 | 158        | 156 | 159 |
| 155 | 158 | <u>154</u> | 156 | 160 |
| 154 | 157 | 158        | 160 | 160 |
| 157 | 157 | 157        | 156 | 155 |

**Step 1:**

$$154 * 4 - 158 - 156 - 158 - 158 = -14$$

So the value after filter = **-14**

We call the resultant image: **sharpened image.**

**Step 2:**

**Filtered image = original + sharpened image**

**The value in the filtered image =  $154 - 14 = 130$**

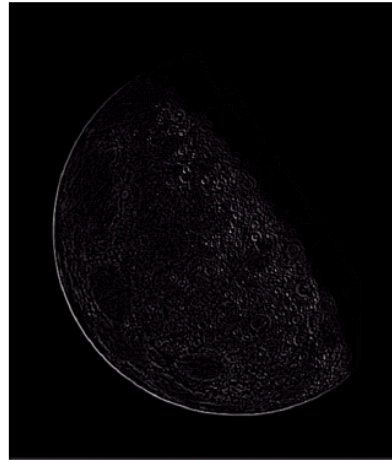
# Laplacian Image Enhancement

|   |    |   |
|---|----|---|
| 0 | 1  | 0 |
| 1 | -4 | 1 |
| 0 | 1  | 0 |



Original  
Image

-



Laplacian  
Filtered Image

=



Sharpened  
Image

In the final sharpened image edges and fine detail are much more obvious

# Simplified Image Enhancement

- The result of a Laplacian filtering is not an enhanced image.
- The entire enhancement can be combined into a **single filtering operation**

$$\begin{aligned} g(x, y) &= f(x, y) - \nabla^2 f \\ &= f(x, y) - [f(x+1, y) + f(x-1, y) \\ &\quad + f(x, y+1) + f(x, y-1) \\ &\quad - 4f(x, y)] \end{aligned}$$

|   |    |   |
|---|----|---|
| 0 | 1  | 0 |
| 1 | -4 | 1 |
| 0 | 1  | 0 |

|       |       |       |
|-------|-------|-------|
| $w_1$ | $w_2$ | $w_3$ |
| $w_4$ | $w_5$ | $w_6$ |
| $w_7$ | $w_8$ | $w_9$ |

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f, & w_5 < 0 \\ f(x, y) + \nabla^2 f, & w_5 > 0 \end{cases}$$

# Simplified Image Enhancement

- ◆ The entire enhancement or sharpening can be done in one PASS.

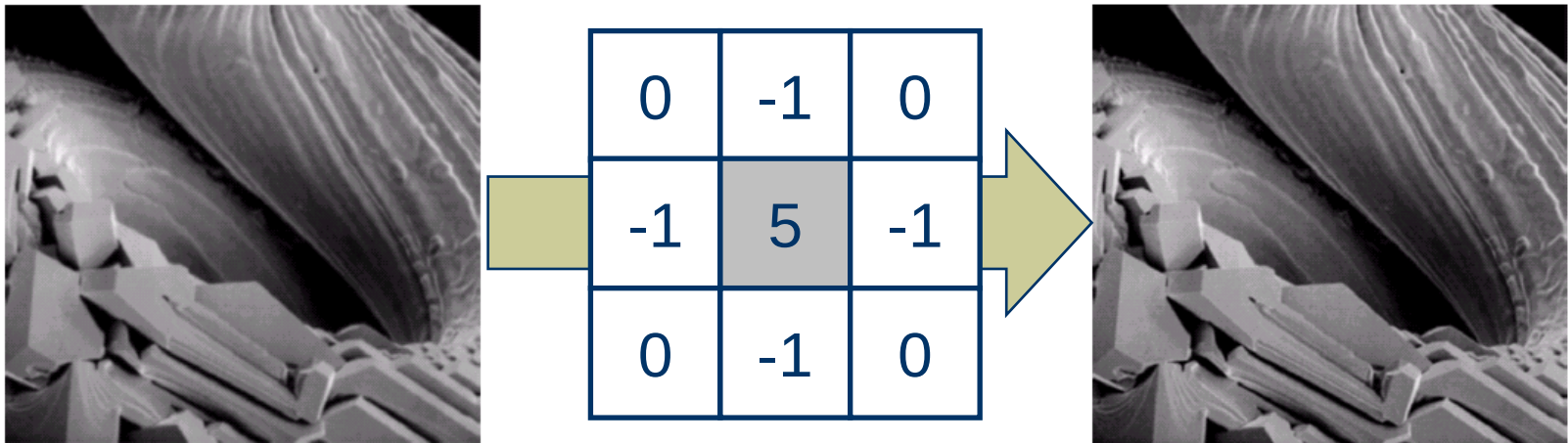
$$\begin{aligned} g(x, y) &= f(x, y) - \nabla^2 f \\ &= 5f(x, y) - f(x+1, y) - f(x-1, y) \\ &\quad - f(x, y+1) - f(x, y-1) \end{aligned}$$

|    |    |    |
|----|----|----|
| 0  | -1 | 0  |
| -1 | 5  | -1 |
| 0  | -1 | 0  |

We can implement it using this filter.

# Simplified Image Enhancement

- ◆ This gives us a new filter which does the whole job for us in one step



# Variants On The Simple Laplacian

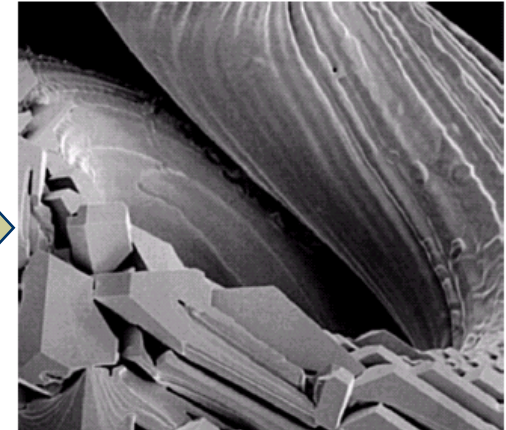
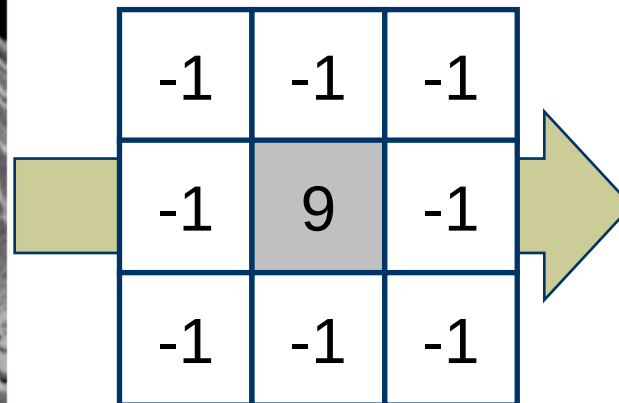
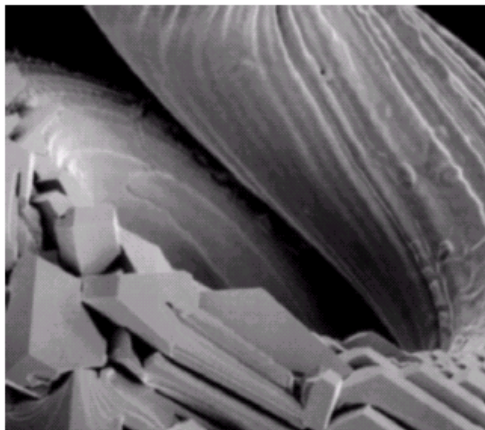
There are lots of slightly different versions of the Laplacian that can be used:

|   |    |   |
|---|----|---|
| 0 | 1  | 0 |
| 1 | -4 | 1 |
| 0 | 1  | 0 |

**Simple  
Laplacian**

|   |    |   |
|---|----|---|
| 1 | 1  | 1 |
| 1 | -8 | 1 |
| 1 | 1  | 1 |

**Variant of  
Laplacian**





# Use of 1st Derivatives for Image Enhancement

The another Sharpening Spatial filters is **SOBEL (Gradient Operators)**.

What is Gradient of a Digital Image?

# The Gradient of a Digital Image

## The Gradient (1<sup>st</sup> order derivative)

- ❑ First Derivatives in image processing are implemented using the magnitude of the gradient.
- ❑ The gradient of function  $f(x,y)$  is

$$\nabla f = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

# The Gradient of a Digital Image

- The magnitude of this vector is given by

$$\text{mag}(\nabla f) = \sqrt{G_x^2 + G_y^2} \approx |G_x| + |G_y|$$

$G_x$

|    |   |
|----|---|
| -1 | 1 |
|----|---|

This mask is simple, and no isotropic. Its result only horizontal and vertical.

$G_y$

|    |
|----|
| 1  |
| -1 |

# The Gradient – First-order Derivative

How can we compute first-order discrete image derivatives?

- There are various ways...
  - One dimensional forward differences
  - Roberts cross gradient operators
  - One dimensional central differences
  - Prewitt operators
  - Sobel operators

There is some debate as to how best to calculate these gradients.

# Gradient Operators

## Robert's Method

- The simplest approximations to a first-order derivative that satisfy the conditions stated in that section are

|       |       |       |
|-------|-------|-------|
| $z_1$ | $z_2$ | $z_3$ |
| $z_4$ | $z_5$ | $z_6$ |
| $z_7$ | $z_8$ | $z_9$ |

$$G_x = (z_9 - z_5) \text{ and } G_y = (z_8 - z_6)$$

$$\nabla f = \sqrt{(z_9 - z_5)^2 + (z_8 - z_6)^2}$$

$$\nabla f \approx |z_9 - z_5| + |z_8 - z_6|$$

# Gradient Operators

- These mask are referred to as the Roberts cross-gradient operators.

|    |   |
|----|---|
| -1 | 0 |
| 0  | 1 |

|   |    |
|---|----|
| 0 | -1 |
| 1 | 0  |

# Sharpening Spatial filters :

## 2) SOBEL (Gradient Operator)

- The Sobel operator provides differencing and smoothing effect of an image.
- Sobel operator consists of 3x3 convolution kernels.  $G_x$  is a simple kernel and  $G_y$  is rotated by  $90^\circ$

|           |           |           |
|-----------|-----------|-----------|
| <b>-1</b> | <b>-2</b> | <b>-1</b> |
| <b>0</b>  | <b>0</b>  | <b>0</b>  |
| <b>1</b>  | <b>2</b>  | <b>1</b>  |

*$G_x$ , Extract horizontal edges*

|           |          |          |
|-----------|----------|----------|
| <b>-1</b> | <b>0</b> | <b>1</b> |
| <b>-2</b> | <b>0</b> | <b>2</b> |
| <b>-1</b> | <b>0</b> | <b>1</b> |

*$G_y$ , Extract vertical edges*

# Gradient Operators

## Sobel Operator

$$\frac{\partial f}{\partial x} = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

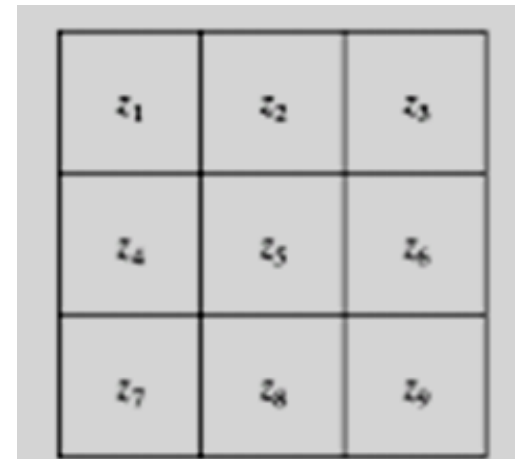
*G<sub>x</sub>, Extract horizontal edges*

$$\frac{\partial f}{\partial y} = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

*G<sub>y</sub>, Extract vertical edges*

$$|G| = |G_x| + |G_y|$$

$$\nabla f \approx \left| (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3) \right| \\ + \left| (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7) \right|$$



*Pixel Arrangement*



# Gradient Operators

## Prewitt Operator

□ is used for detecting edges horizontally and vertically.

$$|G| = |G_x| + |G_y|$$

$$\nabla f \approx |(z_7 + z_8 + z_9) - (z_1 + z_2 + z_3)| \\ + |(z_3 + z_6 + z_9) - (z_1 + z_4 + z_7)|$$

|       |       |       |
|-------|-------|-------|
| $z_1$ | $z_2$ | $z_3$ |
| $z_4$ | $z_5$ | $z_6$ |
| $z_7$ | $z_8$ | $z_9$ |

*Pixel Arrangement*

$$\frac{\partial f}{\partial x} = \begin{array}{|c|c|c|} \hline -1 & -1 & -1 \\ \hline 0 & 0 & 0 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

Extract horizontal edges

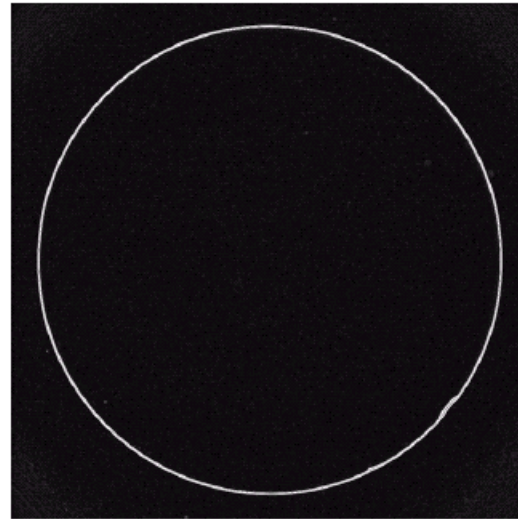
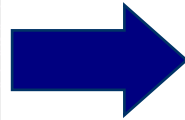
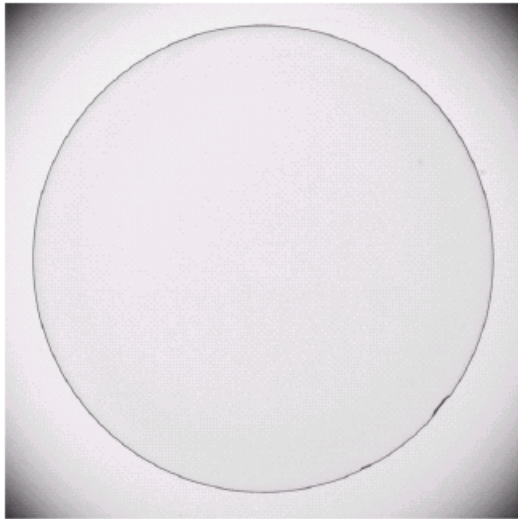
**G<sub>x</sub>**

$$\frac{\partial f}{\partial y} = \begin{array}{|c|c|c|} \hline -1 & 0 & 1 \\ \hline -1 & 0 & 1 \\ \hline -1 & 0 & 1 \\ \hline \end{array}$$

Extract vertical edges

**G<sub>y</sub>**

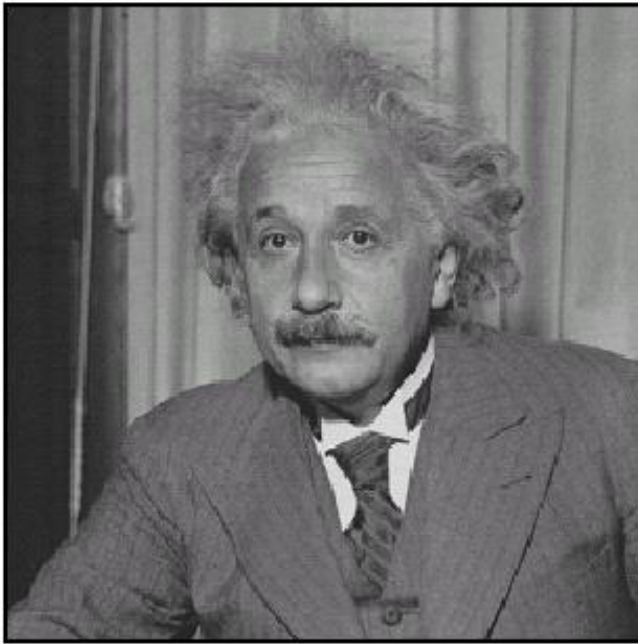
# Sobel Operator: Example



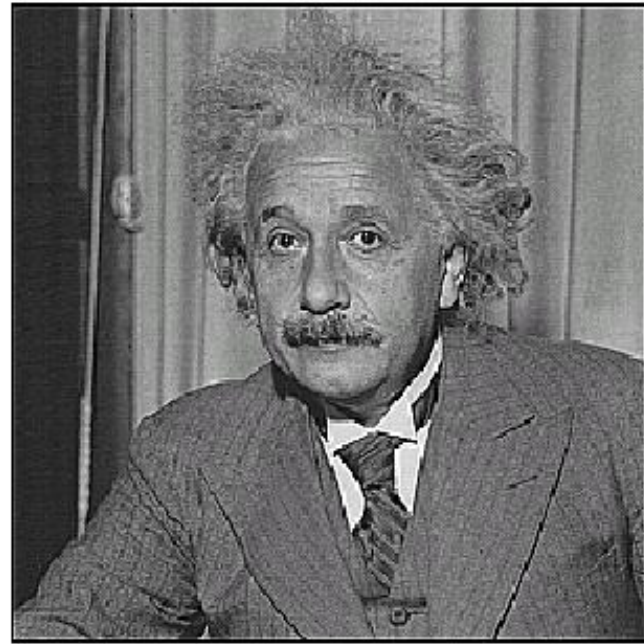
An image of a contact lens which is enhanced in order to make defects more obvious

Sobel filters are typically used for edge detection

# Sharpening with Sobel Operator

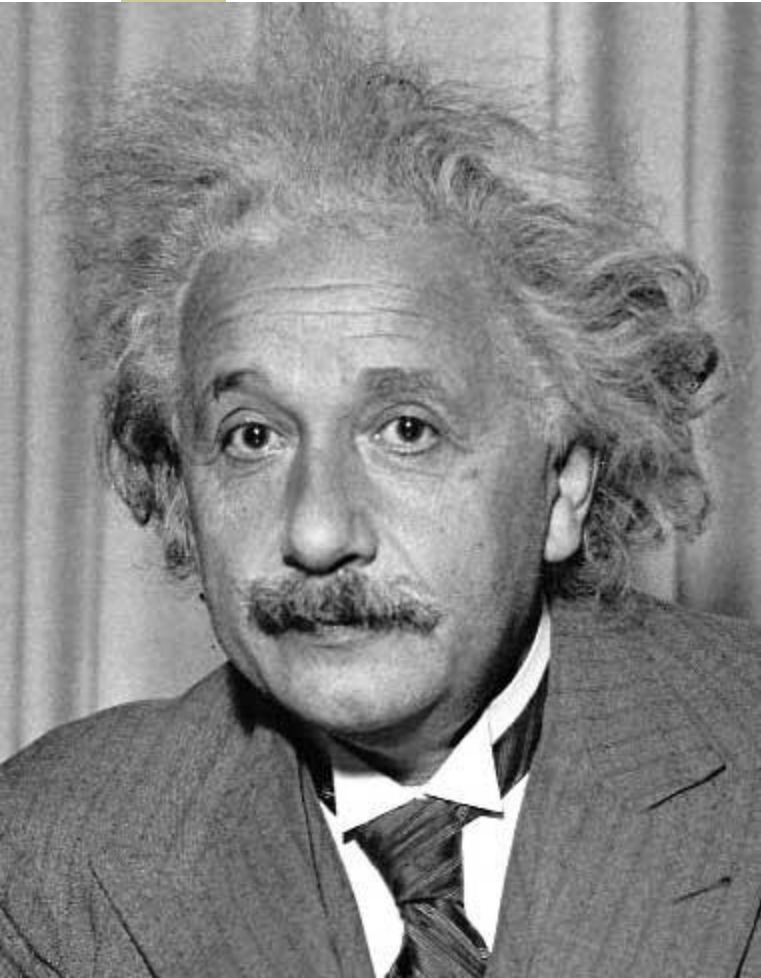


before



after

# Sharpening with Sobel Operator



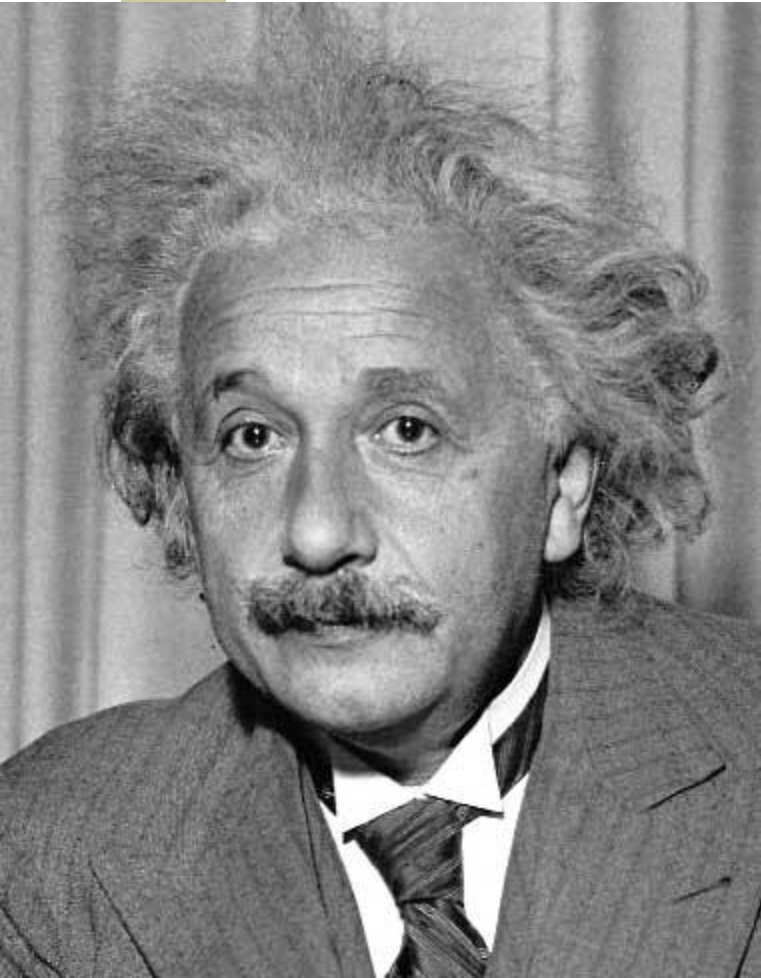
|   |   |    |
|---|---|----|
| 1 | 0 | -1 |
| 2 | 0 | -2 |
| 1 | 0 | -1 |

Sobel



Vertical Edge  
(absolute value)

# Sharpening with Sobel Operator



|    |    |    |
|----|----|----|
| 1  | 2  | 1  |
| 0  | 0  | 0  |
| -1 | -2 | -1 |

Sobel



Horizontal Edge  
(absolute value)



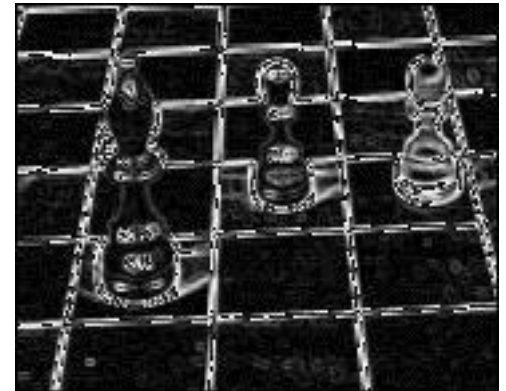
# Sharpening Spatial Filters



Laplacian



Sobel



# Difference Filter

- ❑ Also called as **Emboss filters**
- ❑ Enhances the details in the direction specific to the mask selected
- ❑ Four primary difference filter convolution masks, corresponding to the edges in the vertical, horizontal, and two diagonal directions are:

| Vertical                                                             | Horizontal                                                           | Diagonal 1                                                           | Diagonal 2                                                           |
|----------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|
| $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$ | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$ |

# Difference Filter



Original image



Difference filtered image



Difference filtered image  
added to the original image,  
with contrast enhanced



# Combining Spatial Enhancement Methods

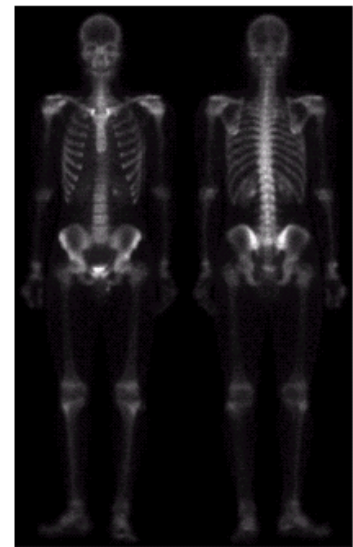
Successful image enhancement is typically not achieved using a single operation

Rather we combine a range of techniques in order to achieve a final result

This example will focus on enhancing the bone scan



# Combining Spatial Enhancement Methods



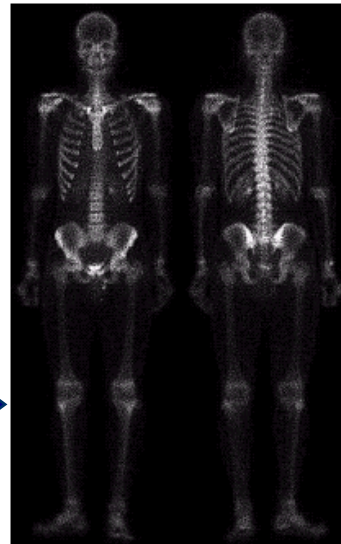
(a)

Laplacian filter of  
bone scan (a)



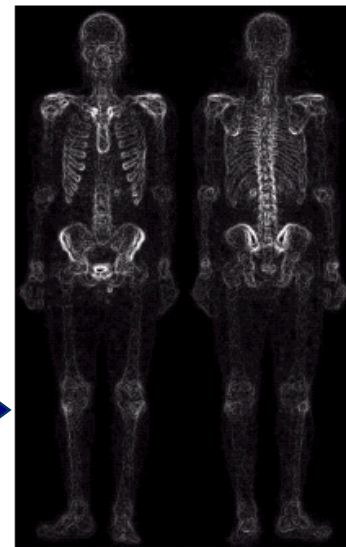
(b)

Sharpened version of  
bone scan achieved  
by subtracting (a)  
and (b)



(c)

Sobel filter of bone  
scan (a)



(d)

# Combining Spatial Enhancement Methods

The product of (c) and (e) which will be used as a mask

(e)

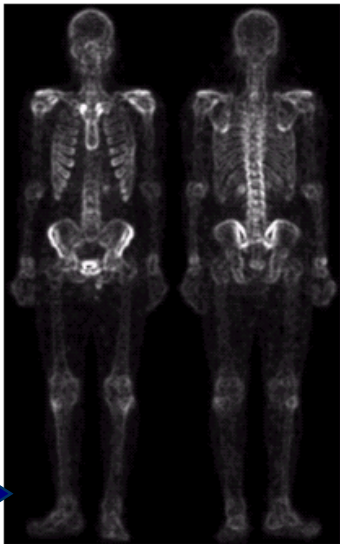
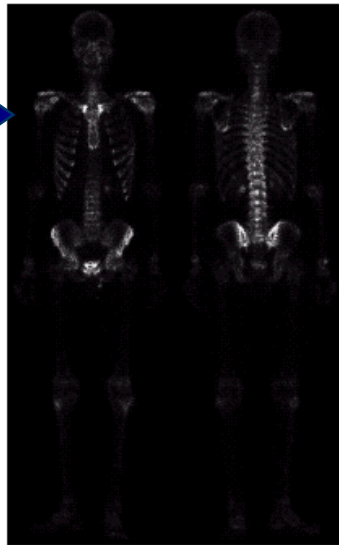


Image (d) smoothed with a 5\*5 averaging filter

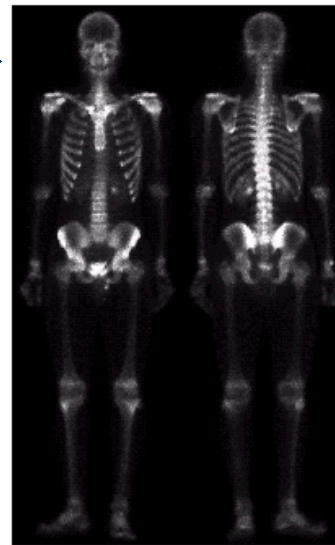
Sharpened image which is sum of (a) and (f)

(f)

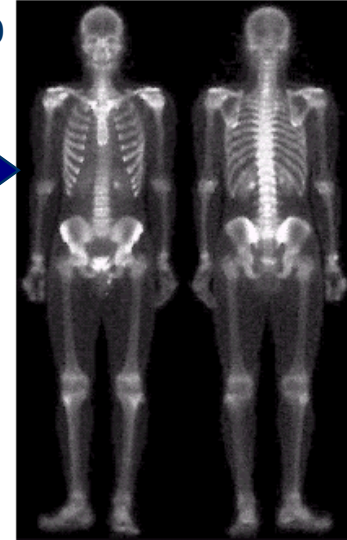


Result of applying a power-law trans. to (g)

(g)



(h)



## Combining Spatial Enhancement Methods

Compare the original and final images



# Class Work

Consider a 3-bit 4x4 image.

|   |   |   |   |
|---|---|---|---|
| 0 | 2 | 6 | 7 |
| 1 | 1 | 6 | 4 |
| 4 | 5 | 2 | 7 |
| 1 | 2 | 6 | 0 |

**Laplacian  
filter**

|    |    |    |
|----|----|----|
| 0  | -1 | 0  |
| -1 | +4 | -1 |
| 0  | -1 | 0  |

Find the filtered output image using

- this **Laplacian** filter,
- a  $3 \times 3$  **Mean** filter
- a  $3 \times 3$  **Median** filter and
- a **Sobel** operator

Ignore the border pixels in calculation and put zero in the border of the output image.